

Questions: Class Resulting Process

Muhammad Usaid
26K-2500

Algorithm

1. Start.
2. Input the numbers of students, N .
3. While $(N < 1)$, ask again.
4. Set Student $\leftarrow 1$
5. While Student $\leq N$:
 1. Set Total $\leftarrow 0$, Subject $\leftarrow 1$ and FSubject $\leftarrow 0$
 2. While Subject ≤ 5 :
 - Input MARKS
 - While (MARKS < 0 OR MARKS > 100), ask again.
 - Add MARKS to Total.
 - If (MARKS < 33), Set FSubject $\leftarrow 1$.
 - Increase Subject by 1.
 3. Calculate Average $\leftarrow \text{Total} / 5$.
 4. If (FSubject = 1), set result to "Fail Subject Deficiency".
 5. Otherwise, if (Average ≥ 80), set result to "Distinction".
 6. Otherwise, if (Average ≥ 60), set result to "Pass".
 7. ^{otherwise,} Set result to "Fail".
 8. Display student number, total, average and result.
 9. Increase student by 1.
6. STOP.

Pseudocode

Begin

Let N be user input

While ($N < 5$), do

 print "Enter at least one student"

 Input N

END while

Student $\leftarrow 1$

While (Student $\leq N$), do

~~Total $\leftarrow 0$~~

~~Subject $\leftarrow 1$~~

~~FSubject $\leftarrow 0$~~

~~While (Subject ≤ 5), do~~

~~print "Enter marks~~

Total $\leftarrow 0$

Subject $\leftarrow 1$

FSubject $\leftarrow 0$

While (Subject ≤ 5), do

 print "Enter marks for subject: ", subject

 Input Mark

 While (Mark < 0 OR Mark > 100), do

 print "Marks must be between 0 and 100"

 Input Mark

 END while.

Total $\leftarrow$ Total + Mark

IF (Mark < 33), then

 FSubject $\leftarrow 1$

END IF

Subject $\leftarrow$ Subject + 1

END While

Average $\leftarrow$ Total / 5

IF (F Subject = 1), then

Result $\leftarrow$ "Fail subject Deficiency"

ELSE IF (Average ≥ 80), then

Result $\leftarrow$ "Distinction"

ELSE IF (Average ≥ 60), then

Result $\leftarrow$ "Pass"

ELSE

Result $\leftarrow$ "Fail"

END IF

Display "Student Number: ", Student

Display "Total Marks: ", Total

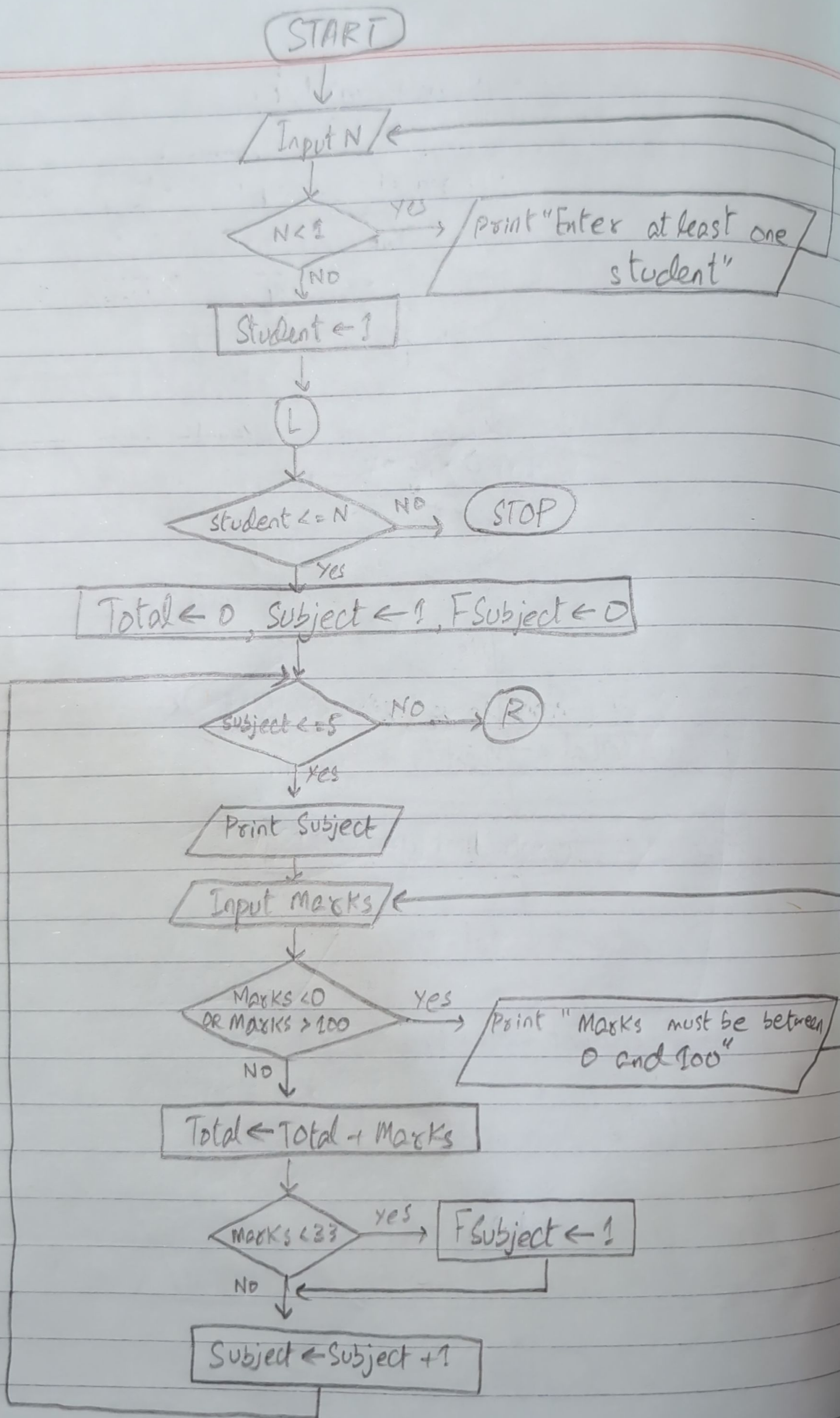
Display "Average: ", Average

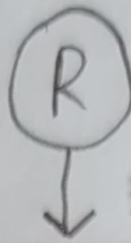
Display "Result: ", Result

Student $\leftarrow$ Student + 1

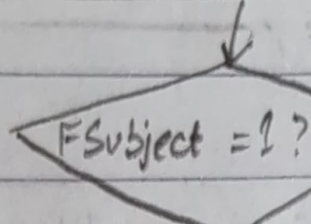
END While

STOP





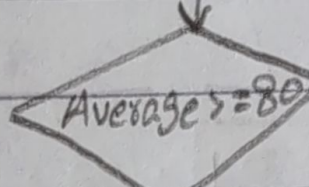
Average $\leftarrow$ Total / 5



yes

Result $\leftarrow$ "Fail Subject Deficiency"

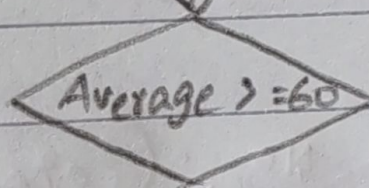
NO



yes

Result $\leftarrow$ "Distinction"

No



yes

Result $\leftarrow$ "Pass"

No

Result $\leftarrow$ "Fail"

Print Student, Total,
Average, Result

Student $\leftarrow$ Student + 1

